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On Chorin's Projection Method for the Incompressible Navier-Stokes Equations

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Summary. Pseudo-compressibility methods are frequently used in computational fluid dynamics in order to cope with the algebraic difficulties caused by the incompressibility constraint. A popular example is the pressure stabilization (Petrov-Galerkin) method of T.J.R. Hughes, et al., which can be applied to the stationary as well as to the nonstationary Navier-Stokes problem. Also the classical projection method of A.J. Chorin can be interpreted as a variant of this method. This observation sheds some new light on the approximation properties of the projection method, particularly for the pressure.

1. Introduction

Let $\Omega \subset \mathbf{R}^n$ $(n = 2,3)$ be a bounded domain with a sufficiently regular boundary $\partial\Omega$. On the cylinder $\Omega\times[0,T]$, we consider the nonstationary Navier-Stokes problem

$$u_t - \nu\Delta u + (u\cdot\nabla)u + \nabla p = f, \quad \nabla\cdot u = 0, \quad \text{in } \Omega\times(0,T], \tag{1.1}$$
$$u_{|\partial\Omega} = 0, \quad u_{|t=0} = u^0.$$

Below, we will use the standard notation $L^2(\Omega)$, $H^k(\Omega)$, and $H_0^k(\Omega)$, for the Lebesgue and Sobolev spaces over Ω, and $(\cdot,\cdot)_\Omega$, $\|\cdot\|_\Omega$ and $\|\cdot\|_{k;\Omega}$ for the corresponding inner products and norms. $L_0^2(\Omega)$ is the subspace of all L^2-functions having mean-value zero. The subscript Ω is usually suppressed. Spaces of vector-valued functions are indicated by bold face type, e.g., $\mathbf{H}_0^1 = [H_0^1]^n$, while no distinction is made in the notation of inner products and norms. Further, the subspaces of solenoidal functions $\mathbf{J}_1 = \{v\in \mathbf{H}_0^1, \nabla\cdot v = 0\}$ and $\mathbf{J}_0 = \{v\in \mathbf{L}^2, \nabla\cdot v = 0$, and $v\cdot n_{|\partial\Omega} = 0$, weakly$\}$ are used. The $\mathbf{L}^2$-projection onto $\mathbf{J}_0$ is denoted by $\mathbf{P}$.

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residui-cau nef $-\Delta p^m = -\frac{1}{k}\nabla\cdot u^m$

$$\int_\Omega \left(\frac{\partial p}{\partial x}\frac{\partial w}{\partial x} + \frac{\partial p}{\partial y}\frac{\partial w}{y}\right) d\Omega + \cancel{\int_{\partial\Omega} \frac{\partial p}{\partial n} w}^{\,0} + \frac{1}{k}\int_\Omega \nabla\cdot u^m \, d\Omega = 0$$

One major computational difficulty in problem (1.1) arises from the incompressibility constraint which gives it a saddle point character. In the earlier days of computational fluid dynamics this has caused severe problems, particularly when a nonstationary flow had to be computed by an implicit time stepping scheme. One of the first approaches to overcome this problem was the so-called "projection method" of Chorin [2], [3] (see also Temam [14]). Here, for $m \geq 1$, one successively computes approximations

$$\text{i)}\quad \tilde{u}^m \in H_0^1: \quad \frac{1}{k}(\tilde{u}^m - u^{m-1}) - \nu\Delta\tilde{u}^m + (\tilde{u}^m\cdot\nabla)\tilde{u}^m = f^m \quad \text{("Burgers step")}, \tag{1.2}$$

$$\text{ii)}\quad u^m \in J_0: \quad u^m = P\tilde{u}^m \quad \text{("projection step")}, \tag{1.3}$$

where k is the time step and $u^0 \in J_1$ is a suitable initial value. The intermediate approximations $\tilde{u}^m$ satisfy the correct boundary condition but violate the incompressibility constraint, while the u^m are divergence free but may have a non-zero tangential component. In view of the orthogonal splitting $L^2 = J_0 \oplus \{\nabla\psi, \psi \in H^1\}$ (see, e.g., [5]), the projection step (ii) can equivalently be written in the form $u^m = \tilde{u}^m - k\nabla p^m$, with some $p^m \in L_0^2 \cap H^1$, which is determined through the Neumann problem

$$\text{ii')}\quad \Delta p^m = \frac{1}{k}\nabla\cdot\tilde{u}^m \text{ in } \Omega, \quad \partial_n p^m|_{\partial\Omega} = 0. \tag{1.4}$$

The formulation (1.4) expresses that the pressure p^m in the projection method automatically satisfy the non-physical boundary condition $\partial_n p^m|_{\partial\Omega} = 0$. This has caused a lot of controversal discussion about the quality of these pressure approximations, which has even led to the believe that these are mere fictitious quantities without any physical relevancy (see, e.g., [15]). The main purpose of this paper is to show that this point of view is wrong. The quantities p^m are indeed proper approximations to the true pressure $p(t_m)$, even in the pointwise sense, in some distance to the boundary $\partial\Omega$. This view is also strongly supported in a recent paper of Gresho, et al., [6]. In fact, the practical success of the projection method has frequently been reported in the literature even for nonstationary flows with complicated vortex structures; see, e.g., Shen [12]. The best known theoretical result is the following error estimate due to Shen [11],

$$\|u^m - u(t_m)\| + \left(k\sum_{\mu=0}^{m}\|p^\mu - p(t_\mu)\|^2\right)^{1/2} = O(\sqrt{k}), \quad 0 < t_m \leq T, \tag{1.5}$$

which is proved for the classical projection method including a first order linearization in the step (i). Furthermore, for a slightly modified scheme Shen [13] obtained the optimal order $O(k)$ in (1.5). Whether this is true also for Chorin's original method was an open question. We will approach this problem through a re-interpretation of the projection method as a certain pressure stabilization method for which a well developped convergence analysis is

available. To this end, we combine the relations (1.2) and (1.4) to obtain that

$$\frac{1}{k}(\tilde{u}^m-\tilde{u}^{m-1}) - \nu\Delta\tilde{u}^m + (\tilde{u}^m\cdot\nabla)\tilde{u}^m + \nabla p^{m-1} = f^m, \quad \tilde{u}^m{}_{|\partial\Omega} = 0, \tag{1.6}$$

$$\nabla\cdot\tilde{u}^m - k\Delta p^m = 0, \quad \partial_n p^m{}_{|\partial\Omega} = 0, \tag{1.7}$$

where $p^0 = 0$. This scheme reminds of a simple pressure stabilization method, with a stabilization parameter $\varepsilon = k$ and an explicit coupling of the pressure in the momentum equation. In this method the quantities $\tilde{u}^m \in H_0^1$ are viewed as the primary approximations to $u(t_m)$. Pseudo-compressibility methods have been frequently used in order to overcome the algebraic difficulties caused by the incompressibility constraint. In this approach additional terms are introduced in the continuity equation in order to stabilize the pressure, e.g.,

$$\nabla\cdot u + \varepsilon p_t = 0, \quad p_{|t=0} = p^0, \qquad \text{(artificial compressibility method)} \tag{1.8}$$

$$\nabla\cdot u + \varepsilon p = 0, \qquad \text{(penalty method)} \tag{1.9}$$

$$\nabla\cdot u - \varepsilon\Delta p = 0, \quad \partial_n p_{|\partial\Omega} = 0. \qquad \text{(Petrov-Galerkin method)} \tag{1.10}$$

Obviously, Chorin's projection method can be interpreted as a pressure stabilization (Petrov-Galerkin) method, in which the pressure term in the momentum equation is treated explicitly and the quantities $\tilde{u}^m \in H_0^1$ are viewed as the primary approximations to $u(t_m)$. As the main result we obtain that Chorin's original projection method is indeed of full first order accurate.

Theorem 1. *The classical projection method* (1.2), (1.3) *admits the error estimates*

$$\|u^m-u(t_m)\|_1 + \|p^m-p(t_m)\| = O(\sqrt{k}), \tag{1.11}$$

$$\|u^m-u(t_m)\| + \|p^m-p(t_m)\|_{-1} = O(k), \tag{1.12}$$

for $0 < t_m \le T$, *where* $\|\cdot\|_{-1}$ *denotes the norm of the dual space of* $L_0^2 \cap H^1$.

The proof of this result will be given below on the basis of a detailed error analysis for the pressure stabilization method. This analysis also indicates that the pressure in the projection method is indeed a reasonable approximation to the true pressure, at least in the interior of the domain Ω. In fact, the effect of the non-physical Neumann boundary condition decays exponentially with rate $\mathrm{dist}(x,\partial\Omega)/\sqrt{\nu k}$, and hence lives only in a narrow boundary layer of width $O(\sqrt{\nu k})$. A simple model example presented in [6] shows an analogous behavior.

2. Pressure Stabilization in the Stationary Stokes Problem

The pressure stabilization (Petrov-Galerkin) method was originally introduced by Hughes, et. al., [9], in the context of finite element discretizations of the steady Stokes and Navier-Stokes problem, in order to enhance the numerical stability properties of these

$$\boxed{\frac{D\vec{u}}{Dt} - \nu \Delta \vec{u} + \overrightarrow{grad}\, p = 0} \quad (1) \quad (3D)$$

aplicando el operador divergencia

$$Div\left(\frac{D\vec{u}}{Dt}\right) - \cancel{Div\left(\nu \Delta \vec{u}\right)}^{\to 0} + div(\overrightarrow{grad}\, p) = 0$$

así considerando "suficiente" continuidad :

$$\frac{D}{Dt}\left(Div\, \vec{u}\right) + \Delta p = 0$$

$$\frac{D}{Dt}\left(\nabla \cdot \vec{u}\right) + \Delta p = 0$$

multiplicando por función test w e integrando

$$\int_\Omega \frac{D}{Dt}(\nabla \cdot \vec{u})\, w + \int_\Omega (\Delta p)\, w = 0$$

aproximamos $\frac{D}{Dt} = \frac{\partial}{\partial t}$

$$+ \int_\Omega (\Delta p)\, w \, d\Omega + \int_\Omega \left(\frac{(\nabla \cdot \vec{u})^{n+1} - (\nabla \cdot \vec{u})^{n}}{\Delta t}\right) w \, d\Omega = 0$$

"equivale sin tener convectivo en (1)"

$$-\int_\Omega (\vec{\text{grad}}\, p \cdot \vec{\text{grad}}\, w)\, d\Omega + \frac{1}{\Delta t}\int_\Omega \left(\frac{(\nabla\cdot\vec{u})^{n+1} - (\nabla\cdot\vec{u})^{n}}{\Delta t}\right) w\, d\Omega = 0$$

con $\boxed{p = -q}$

$$+\int_\Omega \vec{\text{grad}}\, q \cdot \vec{\text{grad}}\, w\, d\Omega - \frac{1}{\Delta t}\int_\Omega \left(\frac{(\nabla\cdot\vec{u})^{n} - (\nabla\cdot\vec{u})^{n+1}}{\Delta t}\right) w\, d\Omega$$

$$\int_\Omega \vec{\text{grad}}\, q \cdot \vec{\text{grad}}\, w - \frac{1}{\Delta t}\int_\Omega \frac{(\nabla\cdot\vec{u})^{n} - \frac{1}{\Omega}\int_\Omega (\nabla\cdot\vec{u})^{n}\, d\Omega}{\Delta t}$$

aproximamos por valor medio

min div en algoritmo de chorin modificado

Resolvemos (1) considerando conocido la presión (luego vemos su cálculo)

$$\int_\Omega \frac{D\vec{u}}{Dt}\, w_i\, d\Omega - \nu\int_\Omega (\Delta\vec{u})\, w_i + \int_\Omega (\vec{\text{grad}}\, p)\, w_i = 0$$

$i = 1,2,3$ (3D)

$$\int \frac{\vec{u}^{n+1} - \overset{=\text{convect}}{u^{n}(\hat{x}, t_n)}}{\Delta t} + \nu\int_\Omega \vec{\text{grad}}\, u \cdot \vec{\text{grad}}\, w + \int_\Omega \text{grad}\, p^{n}\, w = 0$$

lo que está programado en freefem++

Simplificando en (1):

$$\frac{D\vec{u}}{Dt} + \vec{\text{grad}}\, p = 0$$

$$\frac{\vec{u}^{n+1} - \vec{u}^{n}}{\Delta t} = -\vec{\text{grad}}\, p$$

$$\vec{u}^{n+1} = \vec{u}^{n} + \Delta t\, \vec{\text{grad}}\,(-p)$$

$$\boxed{\vec{u}^{n+1} = \vec{u}^{n} + \Delta t\, \vec{\text{grad}}\, q}$$

Finalmente

$$\boxed{q = -p \quad \text{en un proceso iterativo en el cual} \quad p^n \to \overline{p^n} = \frac{\int_\Omega p^n d\Omega}{\text{area}(\Omega)}}$$

es estable

$$\boxed{q = (p^n - p) - \overline{(p^n - q)}}$$

$\equiv$ implementado en freefem

$$\frac{\int_\Omega (p^n - q)\, d\Omega}{\text{area}(\Omega)} = \frac{\int p^n}{\text{area}(\Omega)} - \frac{\int q}{\text{area}(\Omega)} = \overline{p^n} - \overline{q}$$

$$q = p^n - p - \overline{p^n} + \overline{q}$$

$$q = p^n - \overline{p^n} + \overline{q} - p$$

al converger $(p^n - \overline{p^n}) \to 0$, entonces

$q = \bar{q} - p$

dos posibilidades que $p \to 0$

y la otra es la correcta

físicamente

por tanto

$\bar{q} \to 0$

lo que significa que

$\boxed{p = -q}$ como ya habíamos considerado

por tanto el bodeo considerado es la correcta opción!!

```
Vh uold = u, vold = v, pold=p;
Vh f=convect([u,v],-dt,uold), g=convect([u,v],-dt,vold);

solve pb4u(u,w,init=n,solver=LU)
=int2d(Th)(u*w/dt +nu*(dx(u)*dx(w)+dy(u)*dy(w)))
-int2d(Th)((f/dt-dx(p))*w)
+ on(1,u = g0) + on(2,4,u = 0)+ on(3,u=g0);

solve pb4v(v,w,init=n,solver=LU)
= int2d(Th)(v*w/dt +nu*(dx(v)*dx(w)+dy(v)*dy(w)))
-int2d(Th)((g/dt-dy(p))*w)
+on(1,3,v =0)
+on(2,4,v=0);

real meandiv = int2d(Th)(dx(u)+dy(v))/area;

solve pb4p(q,w,init=n,solver=LU)= int2d(Th)
(dx(q)*dx(w)+dy(q)*dy(w))
//- int2d(Th)((dx(u)+ dy(v)-meandiv)*w/dt)+ on(3,q=0);
- int2d(Th)((dx(u)+ dy(v)-meandiv)*w/dt);

real meanpq = int2d(Th)(pold - q)/area;

p = pold-q-meanpq;
u = u + dx(q)*dt;
v = v + dy(q)*dt;
```

$\int \frac{\partial u}{\partial x}$

$\int \frac{\partial p}{\partial y}$

1ª ecuac. movimiento
$p = p_{n+1}$

2ª ecuac. movimiento

corrección porque es $(p^{n+1} - p^n) \to 0$

$$\overline{div\, u} = \frac{\int_{\Omega} div\, u}{area(\Omega)}$$

valor medio de $div\, u$ en todo Ω

p^{n+1}

$u^{n+1} = u^n + \frac{\partial q}{\partial x} \Delta t$